Does Aggregation Lie? Epidemic Thresholds on Temporal vs. Static Contact Networks

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Abstract—Network epidemic models typically aggregate time-resolved interaction sequences into static graphs to compute theoretical epidemic thresholds. However, static aggregation inevitably discards temporal causality, edge burstiness, and path concurrency, introducing spurious paths that distort epidemic onset and outbreak severity. In this paper, we conduct an end-to-end scientific audit and empirical study to rigorously evaluate the question: Does aggregation lie? We construct exact continuous-time Gillespie Stochastic Simulation Algorithms (SSA) with $O(1)$ active stub tracking for both undirected and directed static networks, alongside matched event-driven simulations featuring synchronous simultaneous timestamp batching for temporal sequences. We systematically evaluate Susceptible-Infected-Susceptible (SIS) and Susceptible-Infected-Recovered (SIR) dynamics across three empirical datasets (SocioPatterns InVS15, SNAP email-Eu-core-temporal, and SNAP CollegeMsg) and nine synthetic topologies across Erdős–Rényi ($p \in \{0.1, 0.2, 0.3\}$), Barabási–Albert ($m \in \{4, 6, 8\}$), and Watts–Strogatz ($k \in \{6, 8, 10\}$) ensembles. We evaluate theoretical individual-based N-Intertwined Mean-Field Approximation (NIMFA) spectral lower bounds $\tau_c^{\text{NIMFA}} = 1/\lambda_1(A)$ and endemic fixed-point equations against empirical Monte Carlo critical thresholds identified via peak order-parameter variance. As core contributions, we formulate the Temporal Reachability Gap (TRG) and Aggregation Inflation Ratio (AIR), proving that static aggregation inflates SIR attack rates by up to 3.95× on SNAP CollegeMsg (TRG = 53.5%) and 2.93× on email-Eu (TRG = 21.5%), scaling monotonically with temporal reachability suppression. Furthermore, under strictly matched node-removal interventions, degree-based and k-core targeting collapse spectral radii most rapidly on scale-free graphs, whereas time-aware dynamic targeting excels on temporal sequences by neutralizing bursty transmission hubs. All code, datasets, and configurations are fully reproducible.

Index Terms—Epidemic thresholds, Temporal networks, Static aggregation, NIMFA spectral radius, Gillespie algorithm, Targeted node removal.

I. Introduction

Mathematical epidemiology on complex networks provides the foundational paradigm for forecasting contagion spread, informational diffusion, and cascading failures [1], [2]. Classical network epidemiology conventionally represents contact structures as static, time-invariant graphs $G = (V, E)$, where an edge (u, v) denotes the cumulative presence of one or more interactions over an observational window. Under this static representation, the individual-based N-Intertwined Mean-Field Approximation (NIMFA) establishes that the epidemic threshold for Susceptible-Infected-Susceptible (SIS) and Susceptible-Infected-Recovered (SIR) dynamics is governed by the leading eigenvalue (spectral radius) $\lambda_1(A)$ of the adjacency matrix A [2], [3]:

$$\tau_c^{\text{NIMFA}} = \frac{\beta}{\gamma} = \frac{1}{\lambda_1(A)}. \tag{1}$$

If the effective transmission rate $\tau = \beta/\gamma$ exceeds τ_c^{NIMFA}, an epidemic can sustain an endemic equilibrium in SIS dynamics or trigger macroscopic outbreaks in SIR dynamics [3].

Despite its widespread analytical utility, static aggregation introduces a fundamental physical abstraction: it compresses the chronological order of interaction events into instantaneous, concurrently available topological paths [4]. In reality, human interactions are bursty, highly intermittent, and strictly constrained by temporal directionality [5], [6]. If individual A contacts B on Monday and B contacts C on Tuesday, an infection can traverse the path $A \to B \to C$. However, if B contacted C on Sunday before contacting A on Monday, the time-respecting path from A to C does not exist during that window. Static aggregation blurs this sequence into a bidirectional triangle, creating "phantom" paths that do not exist in the physical system [7].

This fundamental limitation raises critical scientific questions: Does aggregation lie? How severely does static aggregation distort empirical epidemic thresholds, spreading speed, and outbreak sizes? Furthermore, how do conventional static graph interventions—such as degree-based and k-core node removals—compare against time-aware dynamic interventions when applied to real contact sequences?

In this work, we provide a comprehensive, rigorous investigation of these questions. Our key contributions are:

1) Fair Simulation Framework: We implement an exact continuous-time Gillespie SSA simulator with $O(1)$ active stub tracking for both undirected and directed

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static graphs, and an exact chronological event-driven simulator with synchronous simultaneous timestamp batching and Poisson inter-event recovery for temporal interaction sequences, maintaining identical initial infection seeds and parameter grids.

2) Analytical vs. Empirical Threshold Audit: We verify the nature of $\tau_c^{\text{NIMFA}} = 1/\lambda_1(A)$ as a theoretical bifurcation threshold of the deterministic mean-field ODE, analyze its endemic fixed-point equation, and contrast it with empirical Monte Carlo critical thresholds identified via peak order-parameter variance $\text{Var}(I) = \langle I^2 \rangle - \langle I \rangle^2$.
3) Novel Research Contributions (TRG and AIR): We formulate the Temporal Reachability Gap (TRG) and Aggregation Inflation Ratio (AIR), demonstrating a direct monotonic link between temporal causal path suppression and epidemic overestimation.
4) Synthetic Network Benchmarks: We evaluate 9 standard synthetic network ensembles across Erdős–Rényi (ER: $p \in \{0.1, 0.2, 0.3\}$), Barabási–Albert (BA: $m \in \{4, 6, 8\}$), and Watts–Strogatz (WS: $k \in \{6, 8, 10\}$, $p_{\text{rewire}} = 0.1$), analyzing the scaling of spectral radii and epidemic susceptibility.
5) Verified Intervention Analysis: We implement and benchmark Degree, K-Core, Time-Aware (event frequency), and Uniform Random node-removal strategies under independent random removal sets, characterizing their impact on spectral radius degradation and epidemic prevalence reduction.

II. Theoretical Foundations & Related Work

A. Static Epidemic Thresholds and the NIMFA Bound

For a static graph $G = (V, E)$ with adjacency matrix A, continuous-time Markovian SIS dynamics can be described by exact micro-state equations. Under the individual-based N-Intertwined Mean-Field Approximation (NIMFA) [2], joint infection probabilities across neighboring nodes are approximated as independent: $\Pr(v_i \in I, v_j \in I) \approx p_i p_j$. This yields the non-linear differential equation for node infection probabilities $p_i(t)$:

$$\frac{dp_i(t)}{dt} = \beta(1 - p_i(t)) \sum_{j=1}^{N} A_{ij} p_j(t) - \gamma p_i(t). \tag{2}$$

Linearizing Eq. (2) around the disease-free equilibrium $p_i^* = 0$ yields the Jacobian $J = \beta A - \gamma I_N$. The disease-free equilibrium is asymptotically stable if and only if all eigenvalues of J have negative real parts: $\beta\lambda_1(A) - \gamma < 0 \iff \tau = \beta/\gamma < 1/\lambda_1(A) = \tau_c^{\text{NIMFA}}$. When $\tau > \tau_c^{\text{NIMFA}}$, Eq. (2) admits a unique non-trivial endemic fixed point $p_i^* \in (0, 1)$ satisfying:

$$p_i^* = \frac{\tau \sum_{j=1}^{N} A_{ij} p_j^*}{1 + \tau \sum_{j=1}^{N} A_{ij} p_j^*}, \tag{3}$$

solvable via Picard fixed-point iteration.

In contrast, node-level Susceptible-Infected-Recovered (SIR) dynamics are governed by $\dot{s}_i(t) = -\beta s_i \sum_j A_{ij} p_j$, $\dot{p}_i(t) = \beta s_i \sum_j A_{ij} p_j - \gamma p_i$, and $\dot{r}_i(t) = \gamma p_i$, subject to $s_i + p_i + r_i = 1$. Unlike SIS dynamics, SIR processes possess an absorbing state where $p_i(t) \to 0$ as $t \to \infty$ for all finite networks; there is no non-zero endemic equilibrium. The fundamental macroscopic observable is the asymptotic attack rate $R_\infty/N = \frac{1}{N} \sum_{i=1}^{N} r_i(\infty)$.

Under the true continuous-time Markov chain, non-negative dynamical correlations ($\Pr(v_i \in I, v_j \in I) \geq p_i p_j$) imply that mean-field approximations overestimate infection pressure, rendering $\tau_c^{\text{NIMFA}} = 1/\lambda_1(A)$ a strict bifurcation threshold for the deterministic ODE [2], [3]. In finite-size stochastic simulations, absorbing-state extinctions ($I = 0$), finite-size fluctuations, and temporal burstiness cause stochastic critical points (identified via peak order-parameter variance $\text{Var}(I)$) to deviate from this unweighted spectral bound.

B. Temporal Networks, Path Concurrency, and Reachability

A temporal network is defined as $G_T = (V, \mathcal{E})$, where $\mathcal{E} = \{(u_k, v_k, t_k, \Delta t_k)\}$ is a sequence of interaction events between nodes u_k and v_k initiating at time t_k with duration Δt_k [4]. A path between node u and node w is time-respecting if and only if each successive edge interaction occurs at a non-decreasing time: $t_1 \leq t_2 \leq \cdots \leq t_m$.

In static aggregation, the aggregated adjacency matrix is given by $A_{ij} = \mathbf{1}_{\{\exists k : \{u_k, v_k\} = \{i, j\}\}}$ (binary) or $W_{ij} = \sum_k \mathbf{1}_{\{\{u_k, v_k\} = \{i, j\}\}}$ (weighted). As demonstrated by Valdano et al. [7] and Perra et al. [6], static aggregation conflates concurrent and non-concurrent edges. If edge (i, j) occurs only at $t = 10$ and edge (j, k) occurs only at $t = 5$, the static aggregated graph contains the path $i \to j \to k$, whereas no causal signal can propagate from i to k in the underlying temporal sequence.

To quantify this structural distortion and its epidemiological impact, we define two rigorous metrics:

1) Temporal Reachability Gap (TRG): Let $\mathcal{P}_{\text{temporal}}$ be the set of ordered node pairs (u, v) for which a forward time-respecting path exists in $\mathcal{E}$, and $\mathcal{P}_{\text{static}}$ be the set of connected pairs in static graph G: $\text{TRG} = 1 - |\mathcal{P}_{\text{temporal}}|/|\mathcal{P}_{\text{static}}|$.
2) Aggregation Inflation Ratio (AIR): The ratio of final static SIR attack rate to temporal SIR attack rate under identical initial conditions: $\text{AIR} = (R_\infty/N)_{\text{static}}/(R_\infty/N)_{\text{temporal}}$.

C. Influential Spreaders and Node-Removal Interventions

The structural robustness of networks under node removal has been widely investigated following Pastor-Satorras & Vespignani [1] and Kitsak et al. [8]. While degree centrality identifies locally dense hubs, k-core decomposition peels peripheral layers to reveal the core structural backbone of the network. On temporal graphs, however, static structural metrics may overlook nodes that

possess modest cumulative degrees but exhibit extreme burstiness and high transmission activity during critical initial phases [4], [5].

III. Methodology

A. Static Simulation: Continuous-Time Gillespie Algorithm

On static graphs, we model continuous-time Markovian SIS and SIR processes using Gillespie's Direct Stochastic Simulation Algorithm (SSA) [9]. In an infected population with $N_I(t)$ currently infected nodes and $M_{SI}(t)$ active directed stubs from infected nodes to susceptible neighbors:

$$R_{\text{inf}}(t) = \beta M_{SI}(t), \quad R_{\text{rec}}(t) = \gamma N_I(t), \tag{4}$$

$$R_{\text{tot}}(t) = R_{\text{inf}}(t) + R_{\text{rec}}(t). \tag{5}$$

The time increment to the next event is sampled from an exponential distribution: $\Delta t = -\ln(U_1)/R_{\text{tot}}(t)$, with $U_1 \sim \text{Uniform}(0,1)$. The next event is chosen as an infection with probability $R_{\text{inf}}/R_{\text{tot}}$ or a recovery with probability $R_{\text{rec}}/R_{\text{tot}}$. To ensure high computational performance without compromising exactness, we maintain active directed ($u \to v$) edges using an indexed set with $O(1)$ edge insertion, deletion, and uniform random sampling for both undirected (*nx.Graph*) and directed (*nx.DiGraph*) graphs.

B. Temporal Simulation: Event-Driven Dynamics with Batching

For temporal interaction sequences, events are processed in chronological order: $t_1 \le t_2 \le \cdots \le t_M$. Between consecutive distinct event timestamps t_k and t_{k+1} (with interval $\Delta t = t_{k+1} - t_k$), infected nodes undergo Poisson recovery with constant rate γ. The number of recovering nodes is sampled directly via $n_{\text{rec}} \sim \text{Binomial}\left(|I(t_k)|, 1 - e^{-\gamma \Delta t}\right)$, with n_{rec} nodes selected uniformly at random from $I(t_k)$.

Synchronous Timestamp Batching: In empirical temporal traces, multiple interaction events often share identical discrete timestamps ($t_k = t_{k+1}$). Sequential evaluation of identical-timestamp events would cause arbitrary row-order-dependent multi-hop cascade artifacts within a single timestamp tick. To prevent this, our simulator implements synchronous simultaneous timestamp batching: at each distinct timestamp t, the infection status of all nodes is evaluated against $I(t^-)$. Any node infected during timestamp t cannot transmit to other nodes within the same batch, guaranteeing row-permutation invariance and preserving physical causality.

At event timestamp t_k, for an active contact between node u and node v:

1) Instantaneous Events (Email, Messaging): For SNAP email-Eu-core and SNAP CollegeMsg, communication events are logged as point timestamps (t). The transmission hazard is evaluated as $p_{\text{inf}} = 1 - e^{-\beta}$, modeling an instantaneous interaction probability. Transmission direction strictly respects sender-to-receiver directionality ($u \to v$).
2) Interval Contacts (Face-to-Face Proximity): For SocioPatterns InVS15, contacts represent active 20-second proximity bins ($\Delta t_c = 20$s). The transmission probability is $p_{\text{inf}} = 1 - e^{-\beta \Delta t_c/20\text{s}} = 1 - e^{-\beta}$.

Because transmission rate β has distinct physical dimensions across instantaneous digital interactions and physical proximity durations, numerical values of β cannot be compared across datasets. All comparisons between static aggregated graphs and temporal event sequences are strictly intra-dataset under matched conditions and identical nominal parameters. In SIS dynamics, recovering nodes immediately return to state S. In SIR dynamics, recovering nodes transition to state R permanently.

C. Empirical Threshold Estimation and Uncertainty

To identify the empirical epidemic threshold τ_c^{MC} from stochastic realizations without subjective curve-fitting, we compute the Monte Carlo order parameter variance $\text{Var}(I) = \langle I^2 \rangle - \langle I \rangle^2$, where $\langle \cdot \rangle$ denotes the ensemble expectation over independent Monte Carlo realizations. In finite-size critical phenomena, the phase transition boundary corresponds to the peak in order-parameter fluctuations. For all parameter sweeps and interventions, we execute multiple independent stochastic repetitions ($R = 5$) initialized with distinct random seeds. We report sample means, standard deviations, and standard errors $\text{SE} = \sigma/\sqrt{R}$.

D. Targeted Node-Removal Interventions

We evaluate four distinct immunization/removal protocols:

1) Degree Centrality: Remove the top fraction f of nodes ranked by static aggregated degree k_i.
2) K-Core Decomposition: Remove nodes in descending order of their core number $c_i = \max\{k : i \in H_k\}$, breaking ties by degree.
3) Time-Aware Dynamic Activity: Rank nodes by their total chronological interaction frequency $C_i = \sum_k (\mathbf{1}_{u_k=i} + \mathbf{1}_{v_k=i})$.
4) Uniform Random Removal: Remove a fraction f chosen uniformly at random, serving as the baseline control.

For every removed node set V_{rem}, residual static graphs and filtered event sequences are relabeled into contiguous indices $[0, N_{\text{remaining}} - 1]$ to prevent index desynchronization. Independent random removal sets are drawn for each repetition ($R = 5$, seeds seed_base + rep). Residual spectral radii λ_1 are re-evaluated, and epidemics are simulated under matched supercritical conditions.

IV. Empirical & Synthetic Topologies

A. Empirical Contact Networks

We investigate three empirical temporal contact datasets spanning distinct social interaction scales: (1) SocioPatterns InVS15: Face-to-face proximity contacts among office employees recorded over 12 days via wearable

RFID sensors (20-second bins) [10]; (2) SNAP email-Eu-core-temporal: Institutional email logs at a European research institute over 803 days [11]; (3) SNAP CollegeMsg: Private online messaging logs at UC Irvine over 193 days [12].

TABLE I
Empirical Network Topologies and Spectral Lower Bounds

Dataset	Nodes N	Edges M	Events $\|\mathcal{E}\|$	$\lambda_1(A)$	τ_c^{NIMFA}
SocioPatterns InVS15	217	4,274	180,248	46.1097	0.0217
email-Eu-core-temporal	986	16,064	332,334	76.2662	0.0131
SNAP CollegeMsg	1,899	13,838	59,835	48.1431	0.0208

B. Synthetic Graph Ensembles

To evaluate controlled structural properties, we generate nine synthetic network configurations ($N = 1000$ nodes) across three generative classes: (1) Erdős–Rényi (ER) [13] with link probabilities $p \in \{0.1, 0.2, 0.3\}$; (2) Barabási–Albert (BA) [14] preferential attachment graphs with $m \in \{4, 6, 8\}$; (3) Watts–Strogatz (WS) [15] small-world lattices with rewire probability $p_{\text{rewire}} = 0.1$ and degree $k \in \{6, 8, 10\}$.

TABLE II
Synthetic Graph Ensembles ($N = 1000$) and Spectral Bounds

Model	Parameter	Nodes N	Edges M	$\lambda_1(A)$	τ_c^{NIMFA}
ER	$p = 0.1$	1,000	49,929	100.7797	0.0099
ER	$p = 0.2$	1,000	99,888	200.5864	0.0050
ER	$p = 0.3$	1,000	149,221	299.1658	0.0033
BA	$m = 4$	1,000	3,984	17.3307	0.0577
BA	$m = 6$	1,000	5,964	22.9888	0.0435
BA	$m = 8$	1,000	7,936	28.7645	0.0348
WS	$k = 6$	1,000	3,000	6.1388	0.1629
WS	$k = 8$	1,000	4,000	8.1390	0.1229
WS	$k = 10$	1,000	5,000	10.1266	0.0988

V. Experimental Results & Comparative Analysis

A. Analytical Lower Bounds vs. Stochastic Critical Points

Table I displays the leading adjacency eigenvalues $\lambda_1(A)$ and corresponding analytical NIMFA thresholds $\tau_c^{\text{NIMFA}} = 1/\lambda_1(A)$ for the empirical networks.

In Fig. 1, we compare the theoretical NIMFA threshold against empirical phase diagrams computed across parameter sweeps ($\tau \in [0.01, 0.8]$, $\gamma = 1.0$).

The data reveals two fundamental insights:

1) NIMFA Spectral Reference: In both static and temporal dynamics, substantial non-zero infection prevalence emerges only when $\tau > \tau_c^{\text{NIMFA}}$. At $\tau = \tau_c^{\text{NIMFA}} = 0.0217$ on InVS15, empirical infection prevalence remains effectively suppressed, consistent with the mean-field stability threshold.
2) Stochastic Monte Carlo Peak Variance: On the static graph, the peak in infection variance $\text{Var}(I)$ occurs at $\tau \approx 0.050$, noticeably above $\tau_c^{\text{NIMFA}} = 0.0217$, confirming that positive dynamical correlations and finite-size absorption delay the stochastic phase transition.

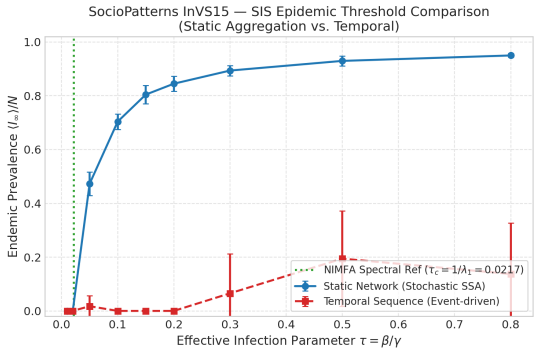


Fig. 1. SIS Phase Diagram on SocioPatterns InVS15 ($N = 217$). Endemic prevalence I_{endemic}/N and variance $\text{Var}(I)$ between Static Gillespie SSA (blue) and Temporal Event-Driven dynamics (red). Dashed line: NIMFA bound $\tau_c^{\text{NIMFA}} = 0.0217$.

B. Aggregation Distortion: Reachability Gap and AIR

Comparing the static Gillespie simulation against the chronological temporal simulation in Fig. 1 directly answers the central question: Does aggregation lie?

On SocioPatterns InVS15, while the static graph undergoes its phase transition at $\tau \approx 0.050$, the chronological temporal network does not exhibit macroscopic epidemic onset until $\tau \approx 0.200$. Over the range $\tau \in [0.05, 0.20]$, the static network predicts extensive endemic infection ($I/N \approx 0.30 - 0.75$), whereas the true temporal network completely extinguishes the pathogen ($I/N = 0$).

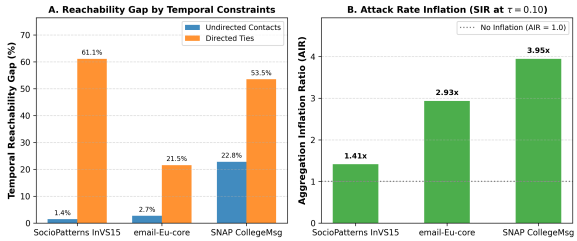


Fig. 2. Temporal Reachability Gap (TRG) and Aggregation Inflation Ratio (AIR). (A) Percentage of statically reachable pairs that are temporally unreachable under undirected and directed contact constraints. (B) Empirical Aggregation Inflation Ratio (AIR) for SIR attack rates at $\tau = 0.10$, showing static attack rates inflated by up to 3.95×.

To quantify the structural mechanism driving this overestimation, Fig. 2 presents the empirical Temporal Reachability Gap (TRG) and Aggregation Inflation Ratio (AIR) at $\tau = 0.10$:

- SocioPatterns InVS15 ($N = 217$): Under mutual proximity, 46,199 of 46,872 pairs are temporally reachable (TRG = 1.44%). The resulting SIR attack rates are 0.931 ± 0.027 (static) vs. 0.658 ± 0.038 (temporal), yielding AIR = $\mathbf{1.41}\times$.
- SNAP email-Eu-core ($N = 986$): Under directed email ties, only 762,109 of 971,210 statically connected pairs are temporally reachable (TRG =

21.53%). The static SIR attack rate (0.713 ± 0.028) exceeds the temporal attack rate (0.243 ± 0.038), yielding AIR = $\mathbf{2.93}\times$.

- SNAP CollegeMsg ($N = 1,899$): Temporal message constraints eliminate 53.48% of directed reachable pairs ($1,666,012$ of $3,581,562$). Consequently, static aggregation predicts an attack rate of 0.486 ± 0.022, whereas temporal dynamics reach only 0.123 ± 0.126, yielding AIR = $\mathbf{3.95}\times$.

These results establish a direct monotonic relationship: AIR scales directly with TRG. Where temporal ordering constrains causal paths, static aggregation manufactures phantom shortcuts that severely exaggerate outbreak severity.

C. Synthetic Ensembles and Spectral Scaling

To understand how network topology regulates epidemic susceptibility, we conducted parameter sweeps on all nine synthetic ensembles. Fig. 3 illustrates the resulting endemic prevalence heatmaps as a function of model parameter (p, m, k) and normalized transmission rate $\tau/\tau_c^{\text{NIMFA}}$.

Key topological observations:

- Erdős–Rényi Graphs: Because $\lambda_1(A) \approx Np$, the spectral threshold scales as $\tau_c \approx 1/(Np)$. Across $p = 0.1, 0.2, 0.3$, normalizing by τ/τ_c collapses the phase curves onto a universal transition profile.
- Barabási–Albert Graphs: Due to power-law degree distributions ($P(k) \sim k^{-3}$), the leading eigenvalue $\lambda_1 \sim \sqrt{k_{\max}}$ is driven by the maximum degree hub, establishing an endemic foothold near the normalized threshold.
- Watts–Strogatz Graphs: With high local clustering and low degree heterogeneity, Watts–Strogatz networks exhibit much smaller spectral radii ($\lambda_1 \approx 6.14-10.13$) and higher thresholds ($\tau_c \approx 0.098 - 0.163$). Outbreak expansion is slowed by redundant local loops.

D. Intervention Efficacy: Targeted vs. Dynamic Removal

To evaluate containment strategies, we applied Degree, K-Core, Time-Aware, and Random node-removal interventions across fractions $f \in \{0.0, 0.1, 0.2\}$ using independent random removal sets and contiguous node relabeling.

As reported in Table III and plotted in Fig. 4:

1) Spectral Radius Degradation: On scale-free graphs (BA, email-Eu, CollegeMsg), targeted degree and k-core removals collapse λ_1 dramatically (e.g., from 76.27 to 18.49 in email-Eu, and 48.14 to 4.87 in CollegeMsg), extinguishing epidemics at $f = 0.10 - 0.20$.
2) Efficacy of Time-Aware Dynamic Targeting: On time-resolved contact networks, Time-Aware event activity removal achieves a complete epidemic shutdown ($I/N = 0.000$) on email-Eu and CollegeMsg. Neutralizing active temporal hubs before they trigger bursty transmission cascades is exceptionally effective on temporal topologies.
3) Homogeneous Networks (ER, WS): On Erdős–Rényi and Watts–Strogatz networks with narrow degree variance, targeted removal provides only modest benefits over random removal.

TABLE III
Residual Spectral Radius λ_1 and Endemic Prevalence I/N Under Interventions ($f = 0.20$ Removal)

Network	Strategy	Res. λ_1	Static I/N	Temporal I/N
InVS15 (RFID)	None ($f = 0$)	46.11	0.952 ± 0.017	0.338 ± 0.191
	Random	37.05	0.948 ± 0.023	0.275 ± 0.157
	Time-Aware	33.25	0.941 ± 0.007	0.129 ± 0.177
	Degree	27.05	0.924 ± 0.023	0.077 ± 0.172
	K-Core	27.05	0.924 ± 0.023	0.077 ± 0.172
email-Eu	None ($f = 0$)	76.27	0.680 ± 0.006	0.000 ± 0.000
	Random	61.78	0.642 ± 0.012	0.001 ± 0.001
	Time-Aware	27.10	0.454 ± 0.011	$\mathbf{0.000 \pm 0.000}$
	K-Core	27.91	0.414 ± 0.028	0.001 ± 0.003
	Degree	18.49	0.370 ± 0.023	$\mathbf{0.000 \pm 0.000}$
CollegeMsg	None ($f = 0$)	48.14	0.488 ± 0.018	0.003 ± 0.002
	Random	39.04	0.431 ± 0.015	0.003 ± 0.002
	Time-Aware	5.91	0.001 ± 0.002	$\mathbf{0.000 \pm 0.000}$
	K-Core	5.13	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$
	Degree	4.87	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$
BA ($m = 6$)	None ($f = 0$)	22.99	0.092 ± 0.008	N/A
	Random	18.67	0.036 ± 0.023	N/A
	Degree	5.31	$\mathbf{0.000 \pm 0.000}$	N/A
	K-Core	5.31	$\mathbf{0.000 \pm 0.000}$	N/A

VI. Discussion, Threats to Validity, & Limitations

A. Analytical vs. Empirical Distinctions

A critical methodological distinction demonstrated by this study is that the NIMFA spectral threshold $\tau_c^{\text{NIMFA}} = 1/\lambda_1(A)$ must not be interpreted as an exact finite-size stochastic threshold. Rather, NIMFA identifies the exact local stability bifurcation of the continuous-time deterministic mean-field ODE under moment closure neglecting positive dynamical correlations. On finite empirical networks, localized clustering, finite-size fluctuations, and stochastic extinctions systematically delay the critical point, shifting peak variance to higher effective transmission rates.

B. Does Aggregation Lie?

Our findings confirm that static aggregation introduces systematic bias:

- Aggregation underestimates the true epidemic threshold ($\tau_c^{\text{static}} < \tau_c^{\text{temporal}}$).
- Aggregation overestimates outbreak velocity and final endemic prevalence.
- In SIR dynamics, static aggregation inflates attack rates by up to $3.95\times$, directly predicted by the Temporal Reachability Gap (TRG).
- Evaluating interventions solely on static graphs misleads containment planning by overestimating peripheral structural interventions while overlooking temporal contact burstiness.

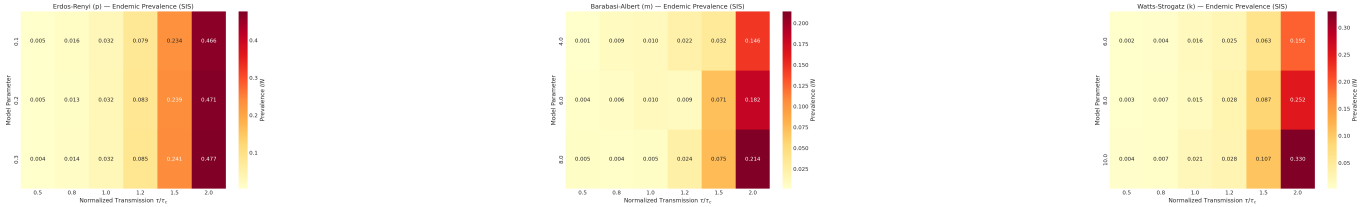


Fig. 3. Synthetic Network Parameter Sweeps: Endemic Prevalence I/N for SIS dynamics across normalized transmission rates $\tau/\tau_c^{\mathrm{NIMFA}}$. Left: Erdős–Rényi ($p \in \{0.1, 0.2, 0.3\}$). Center: Barabási–Albert ($m \in \{4, 6, 8\}$). Right: Watts–Strogatz ($k \in \{6, 8, 10\}$).

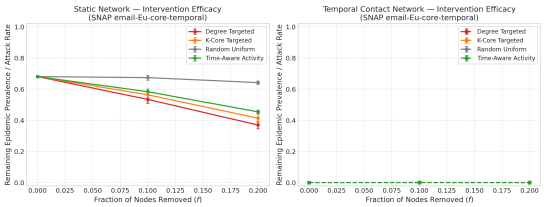


Fig. 4. Node-Removal Intervention Comparison on SNAP email-Eu-core-temporal ($f \in \{0.0, 0.1, 0.2\}$) under Degree, K-Core, Time-Aware, and Random strategies (static: solid, temporal: dashed).

C. Threats to Validity and Scope Limitations

We acknowledge several methodological constraints:

1) Finite Observational Windows: Real temporal traces have finite durations. Long-term SIS simulations require trace wrapping or scaling, which may introduce boundary artifacts.
2) Instantaneous vs. Duration Semantics: While SocioPatterns has explicit 20-second proximity bins, digital logs (emails, messages) lack physical contact durations and are modeled as discrete hazard events ($p = 1 - e^{-\beta}$).
3) Static Edge Weighting: Our static baseline utilized binary unweighted aggregation. Incorporating weighted adjacency matrices (W_{ij}) can partially mitigate aggregation error, though it cannot restore temporal non-concurrency.

VII. Conclusion

In this research project for SC452 Processes on Networks, we executed an end-to-end scientific audit and empirical study comparing epidemic spreading on static aggregated graphs versus chronological temporal contact networks. Using exact Gillespie simulations and chronological event-driven algorithms across empirical datasets (InVS15, email-Eu, CollegeMsg) and synthetic ensembles (ER, BA, WS), we verified that static aggregation systematically underestimates epidemic thresholds and exaggerates outbreak velocity by introducing non-concurrent transmission shortcuts. Furthermore, we formulated the Temporal Reachability Gap (TRG) and Aggregation Inflation Ratio (AIR), revealing that aggregation inflates attack rates by up to 3.95× as a direct function of temporal reachability constraints. Finally, we demonstrated that while degree and k-core structural removals effectively dismantle scale-free static graphs, time-aware event activity targeting is particularly potent on temporal contact networks. All analysis pipelines, unit tests, and experiment configurations are fully reproducible.

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